VEL SOFTWARE SOLUTIONS · VEL-CSE-001
Companion template to the Custom Software & Enterprise Solutions Department Handbook · v1.0 · 4 November 2026

Appendix L — Project Costing Model Specification

Specification for the department's estimating workbook (spreadsheet or small web tool). Implementation must reproduce the worked examples in Chapter 8 to the rupee — those are the acceptance tests.

Sheets

Sheet	Purpose	Key columns / cells
`Inputs`	Global assumptions (versioned, dated).	Cost/day by role · target margin · PM ratio · phase ratios · contingency by size · overhead/month · management fee % · support/upgrade % · tax note
`RateCard`	Billing rates derived from cost.	Role · cost/day · rate/day = cost ÷ (1 − margin)
`Modules`	Three-point estimates.	Module · FR range · O · M · P · expected = (O+4M+P)/6 · risk flag
`Phases`	Effort by lifecycle component.	Component · basis · days · cost/day · cost
`Integrations`	Per interface.	Name · complexity (3/6/12 d) · vendor · sandbox availability
`Migration`	Per data source.	Source · tables · quality · days
`Hosting&Licences`	Quoted from **official calculators/vendor quotes**.	Service · SKU · region · unit price · qty · months · source URL · date verified · screenshot ref
`Summary`	Price, margin, schedule.	Labour · contingency · licences · hosting · direct · price · margin % · payment schedule
`Recurring`	Annual support, hosting, upgrades.	Support % · hosting × 12 × (1+fee) · upgrade block
`BreakEven`	Department economics.	Overhead/month · contribution per project · projects/year
`ChangeRequests`	CR log and pricing.	CR ID · roles/days · assessment fee · post-freeze premium · status

Formulas (authoritative)

```
LabourCost      = Σ ( Days[c] × CostPerDay[c] )                  over all components c incl. Project Management
PM_Days         = PM_Ratio × Σ Days[c ≠ PM]
Contingency     = LabourCost × Risk%                             (10 % small, 15 % medium, 20 % enterprise)
DirectCost      = LabourCost + Contingency + Licences + BuildHosting
Price           = DirectCost ÷ (1 − TargetMargin)                (TargetMargin default 0.40; floor 0.30)
GrossMargin%    = (Price − DirectCost) ÷ Price
ModulePrice[m]  = (Price − SharedPrice − LicHostPrice) × DevDays[m] ÷ Σ DevDays
SharedPrice     = (IntegrationCost + MigrationCost) × (1 + Risk%) ÷ (1 − TargetMargin)
AnnualSupport   = Support% × Price ;  AnnualHosting = Monthly × 12 × (1 + MgmtFee)
BreakEven       = 12 × OverheadPerMonth ÷ (Price − DirectCost)   projects per year
ChangeRequest   = 0.5×BA_rate + Σ(days × rate) × (1 + 0.20 if after freeze)
```

Controls

- Inputs sheet is **versioned and dated**; changing a rate requires Department Head approval.
- Every price in `Hosting&Licences` carries source URL, SKU, region, date verified — no row without evidence.
- Technical Lead reviews module days; Finance reviews margin; both initial the Summary sheet.
- Model must fail validation if: margin < floor, contingency missing, any of the 17 cost components has zero days without a written reason.
- Export a PDF quote snapshot stored with the SOW; never overwrite a quoted version.

Acceptance tests for the model

Case	Expected (ex-tax)
Small example (Ch 8)	Price ₹6,82,567 · margin ₹2,73,027 (40 %)
Medium example	Price ₹27,08,833 · margin ₹10,83,533
Enterprise example	Price ₹93,87,033 · margin ₹37,54,813
Change request (Medium, 1 BA + 4 Dev + 2 QA + 0.5 DB)	₹64,333

Approval

Role	Name	Signature	Date
Technical Lead			
Finance			
Department Head			